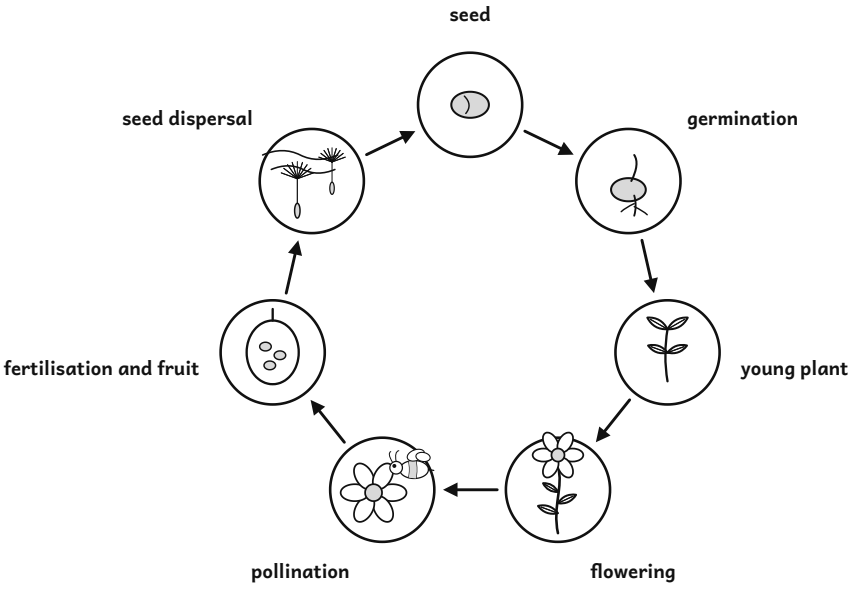
LIVING THINGS

Flowering Plants: Reproduction and Life Cycles

Flowers, pollination, fertilisation, seed dispersal and the life cycle of a flowering plant



In this pack

Learn it: How flowering plants make new plants ★☆☆
Fact check: Parts of a flower ★☆☆
Think it through: From flower to new plant ★★☆
Investigate: Spinning seeds: a paper helicopter test ★★☆
Real-world case study: Bees, mustard and honey ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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How flowering plants make new plants

Name: Class: Date:

Learning goals

- Name the male and female parts of a flower and say what they do.
- Explain pollination, fertilisation and seed dispersal.
- Describe the life cycle of a flowering plant.

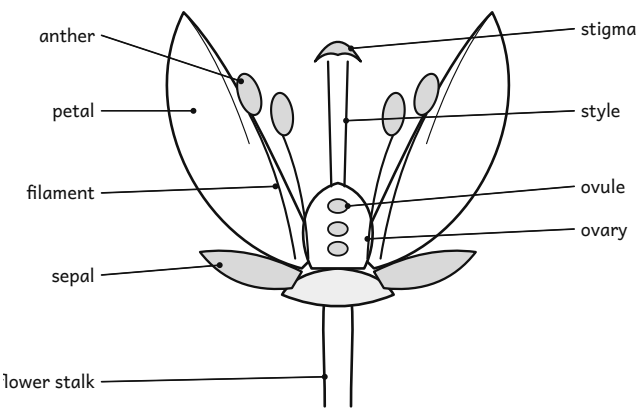
Every spring the mango trees of Rajshahi and Chapainawabganj are covered in tiny flowers called *mukul*. A few months later the same branches are heavy with mangoes. Where did the fruit come from? It came from the flowers. **Flowers are the parts of a plant that make seeds**, and seeds grow into new plants. Making new living things of the same kind is called **reproduction**.

Inside a flower

A flower is like a small factory with different workers. The green **sepals** protect the flower while it is a bud. The **petals** are often brightly coloured and scented to attract insects.

The **stamens** are the *male* parts. Each stamen has a thin stalk called a **filament** holding up an **anther**. The anther makes **pollen**, a fine powder that contains the male cells.

The **carpel** is the *female* part, in the centre of the flower. At the top is the sticky **stigma**, which catches pollen. The **style** is a tube joining the stigma to the **ovary**. Inside the ovary are tiny **ovules**. Each ovule contains a female cell, the egg (ovum).



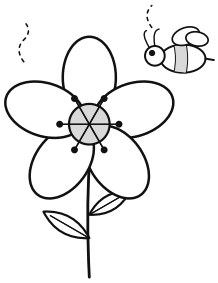
A flower cut in half to show its parts.

Pollination: moving pollen

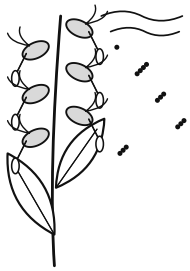
Before a seed can form, pollen must travel from an anther to a stigma. This is called **pollination**. Plants cannot walk, so they need help.

Insect-pollinated flowers have large, bright petals, a sweet scent and sugary **nectar**. When a bee visits to drink nectar, sticky pollen rubs onto its hairy body. At the next flower, some of that pollen brushes off onto the stigma.

Wind-pollinated flowers, such as grasses and maize, are small and dull with no scent or nectar. Their anthers hang outside the flower and release huge amounts of light, dusty pollen. Their feathery stigmas catch pollen floating in the air.



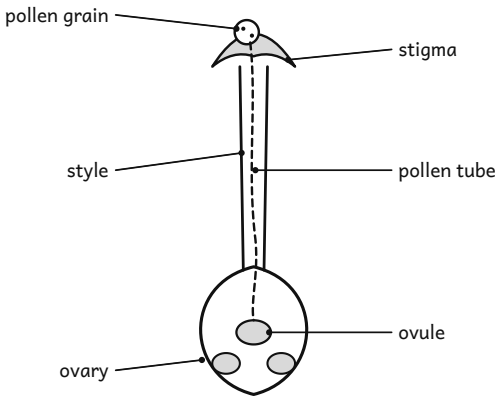
insect-pollinated



wind-pollinated

Fertilisation: a seed begins

When a pollen grain lands on the right kind of stigma, it grows a thin **pollen tube** down through the style into the ovary. The male cell travels down the tube and joins with the female cell inside an ovule. This joining is called **fertilisation**.



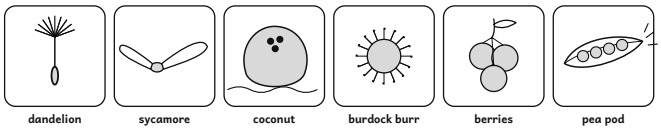
The pollen tube carries the male cell to the ovule.

After fertilisation, the petals wither and fall. Each fertilised **ovule becomes a seed**, and the **ovary grows into a fruit** that protects the seeds. A mango has one large seed; a tomato or a guava has many.

Seed dispersal: spreading out

If all the seeds dropped under the parent plant, the young plants would compete with their parent and with each other for **light, water and space**. So plants spread their seeds away. This is called **seed dispersal**.

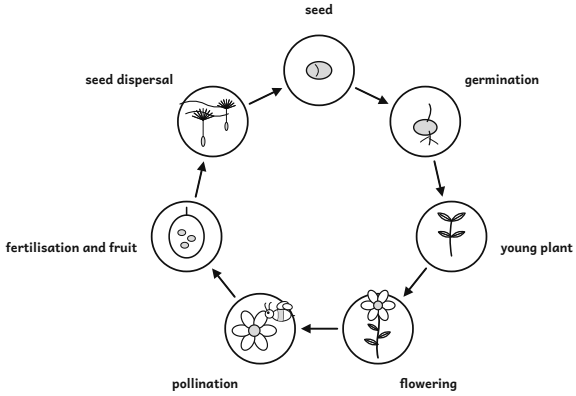
- **Wind:** light seeds with parachutes or wings, such as the fluffy seeds of the *shimul* (silk-cotton) tree, dandelions and sycamore.
- **Water:** seeds that float, such as coconuts and the seedlings of mangrove trees in the Sundarbans.
- **Animals (eaten):** juicy fruits such as guava and berries. Animals eat the fruit and drop the seeds in their droppings far away.
- **Animals (hooks):** burrs with tiny hooks, such as *chorkanta* grass, catch on fur and clothes.
- **Explosion:** pods that dry and burst open, flinging seeds out, such as peas and *dopati* (balsam).



Seeds are shaped for the way they travel.

The life cycle

A **life cycle** is the series of changes a living thing goes through in its life. For a flowering plant it goes round and round: a seed **germinates**, grows into a young plant, then an adult plant that **flowers**. Its flowers are **pollinated**, **fertilisation** happens, fruits and seeds form, the seeds are **dispersed**, and new seeds germinate.



The life cycle of a flowering plant.

Did you know?

Honeybees tell each other where to find flowers by doing a "waggle dance" inside the hive. The angle and length of the dance show the direction and distance of the food.

Bangladesh connection

In winter, the yellow mustard fields of Bangladesh buzz with bees. Many beekeepers move their hives to these fields. The bees collect nectar to make honey and, at the same time, pollinate the mustard so that the plants make more seeds for mustard oil.

Think like a scientist

Plants and animals depend on each other. Bees need flowers for food; many flowers need bees to carry their pollen. If pollinators disappeared, what would happen to the fruits and vegetables we eat?

Key words

reproduction	making new living things of the same kind
stamen	male part of a flower: anther + filament
carpel	female part of a flower: stigma + style + ovary
pollen	powder made by the anther; contains male cells
ovule	part inside the ovary that contains the female cell and becomes a seed
pollination	moving pollen from an anther to a stigma
fertilisation	a male cell from pollen joins with a female cell in an ovule
seed dispersal	spreading seeds away from the parent plant
germination	when a seed starts to grow
nectar	sugary liquid in flowers that attracts animals

Parts of a flower

★☆☆
Grade 5

Name: Class: Date:

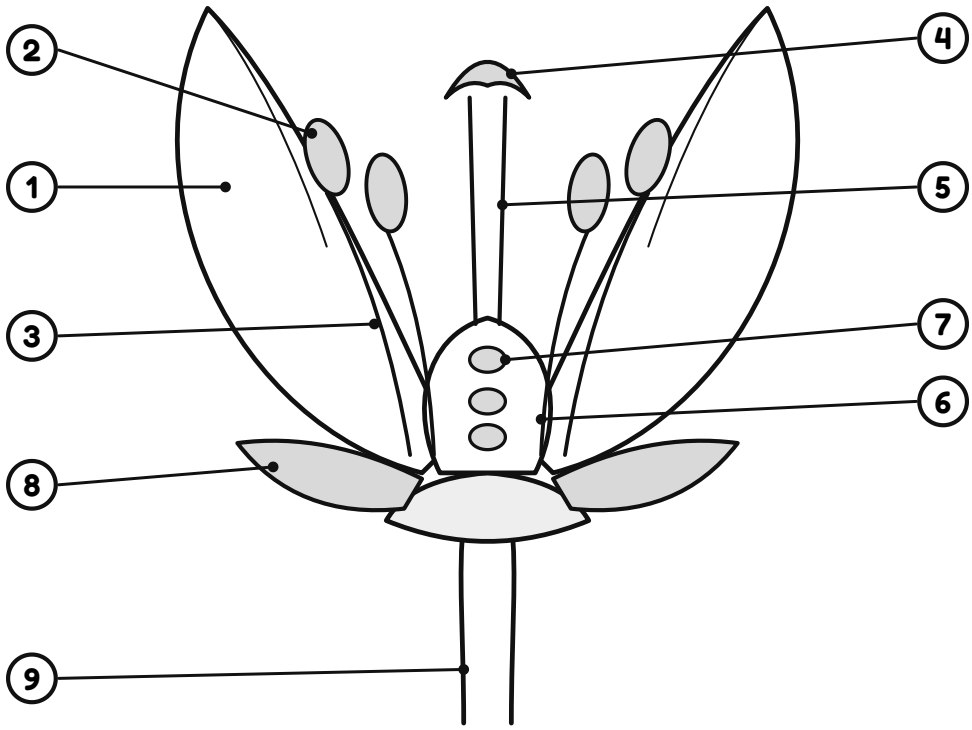
Learning goals

- Label the parts of a flower.
- Sort the parts into male and female.
- Use the key words correctly.

1 **Label the parts of this flower. Use the words in the box.** [9]

LABELS

stigma **ovule** **style** **petal** **flower stalk** **anther** **filament** **ovary** **sepal**



1 _____ 2 _____

3 _____ 4 _____

5 _____ 6 _____

7 _____ 8 _____

9 _____

2 **Sort the flower parts into the table.** [4]

SORT THESE

petal **style** **filament** **anther** **sepal** **stigma** **ovule** **ovary**

Male parts	Female parts	Other parts

3 **Match each word to its meaning.** [6]

Draw a line to match each pair, or write the letter.

pollination ●	● **A** seeds are carried away from the parent plant
fertilisation ●	● **B** a seed starts to grow
seed dispersal ●	● **C** a powder made by the anther
germination ●	● **D** a sweet liquid that attracts insects
nectar ●	● **E** a male cell joins with a female cell in an ovule
pollen ●	● **F** pollen moves from an anther to a stigma

4 **True or false? Tick the box. If a sentence is false, write the correct version.** [5]

Statement	True	False	If false, correct it
1. The anther is a female part of the flower.			
2. Pollen contains male cells.			
3. After fertilisation, the ovary grows into a fruit.			
4. Wind-pollinated flowers usually have big, bright petals.			
5. Each fertilised ovule becomes a seed.			

From flower to new plant

★★☆ Grade 5

Name: Class: Date:

Learning goals

- Put the stages of a life cycle in order.
- Compare insect- and wind-pollinated flowers.
- Explain how seeds are shaped for dispersal.

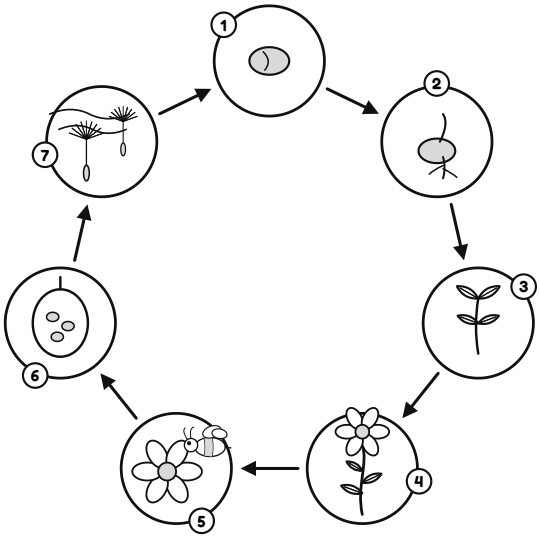
1 **These sentences describe what happens after a bee visits a flower. Number them 1 to 6 in the right order.** [3]

- ☐ The ovule becomes a seed and the ovary becomes a fruit.
- ☐ A bee carries pollen from another flower.
- ☐ The fruit is eaten and the seed is dropped far away.
- ☐ The male cell joins the female cell in the ovule.
- ☐ Pollen lands on the sticky stigma.
- ☐ A pollen tube grows down the style to the ovary.

2 **Label the stages of the life cycle.** [7]

LABELS

fertilisation and fruit **seed dispersal** **pollination** **seed** **flowering** **germination** **young plant**



1 _____ 2 _____

3 _____ 4 _____

5 _____ 6 _____

7 _____

3 **Read the descriptions of flowers A and B.** [4]

Flower A	Flower B
Tiny green flowers, no smell, anthers hanging outside, feathery stigmas, lots of light pollen.	Large purple petals, sweet smell, nectar, sticky pollen.

a) Which flower is pollinated by insects? Give two reasons.

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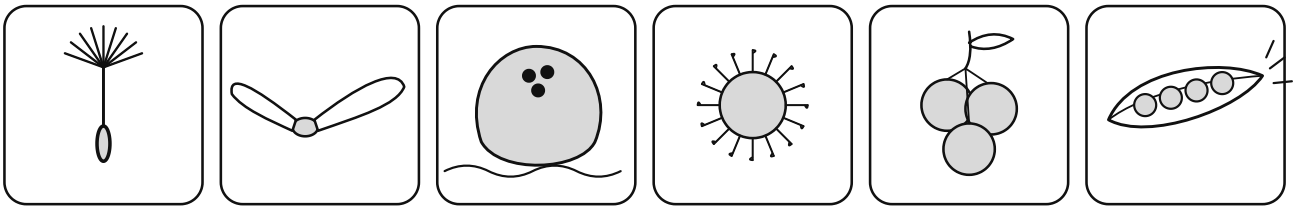
.....

b) Why does flower A make so much pollen?

.....

.....

4 **Look at the seeds A to F. Complete the table.** [6]



A B C D E F

Seed	How is it dispersed?	Which feature helps?
A		
B		
C		
D		
E		
F		

5 **Why is it important for seeds to be dispersed away from the parent plant?** [2]

.....

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.....

INVESTIGATE · A6 · FLOWERING PLANTS

Spinning seeds: a paper helicopter test

★★☆ Grade 5

Name: Class: Date:

Learning goals

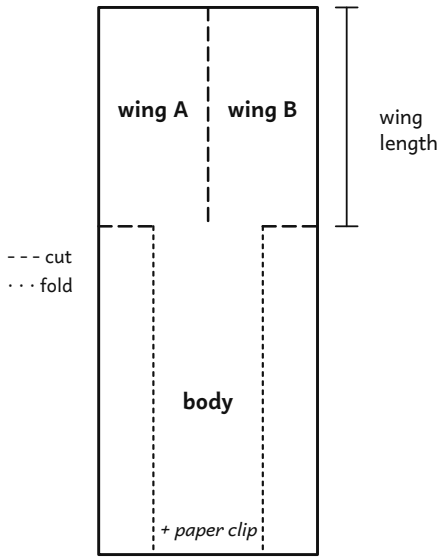
- Use a model to investigate wind dispersal.
- Identify variables in a fair test.
- Calculate averages, spot a result that does not fit and draw a conclusion.

Sycamore and other winged seeds spin as they fall. Spinning slows them down, so the wind has more time to carry them away. Class 5B made paper helicopters as **models** of winged seeds. They asked:

Does wing length change how long a paper helicopter takes to fall?

You will need

paper, scissors, ruler, paper clip, stopwatch, a safe place to drop from 2 m



Paper helicopter template

1 **Name the variables in this investigation.** [3]

Variable	In this investigation
The variable we change (independent)	
The variable we measure (dependent)	
Two variables we keep the same (controlled)	

2 **Here are the class results. Each helicopter was dropped three times.** [5]

Wing length (cm)	Drop 1 (s)	Drop 2 (s)	Drop 3 (s)	Average (s)
4	1.2	1.3	1.1	
6	1.6	1.5	1.7	
8	2.0	2.1	1.9	
10	2.4	1.5	2.6	

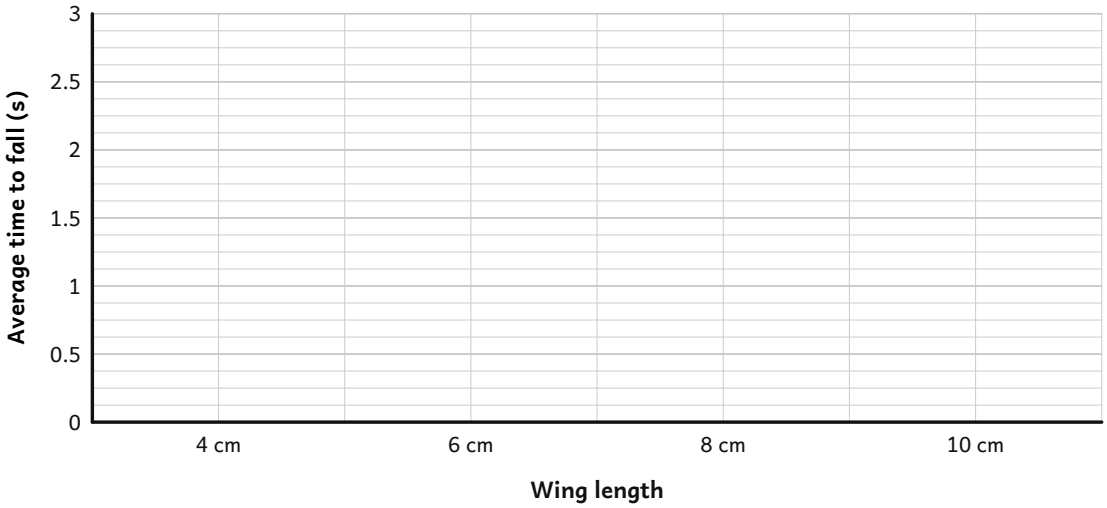
a) Work out the average for each wing length and write it in the table.

b) Which result does not fit the pattern? Suggest why it happened.

.....

.....

3 **Draw a bar chart of the average times.** [4]



4 **Write a conclusion. Use the words *longer* and *slower*.** [2]

.....

.....

5 **How does this help a real sycamore tree?** [2]

.....

.....

REAL-WORLD CASE STUDY · A6 · FLOWERING PLANTS

Bees, mustard and honey

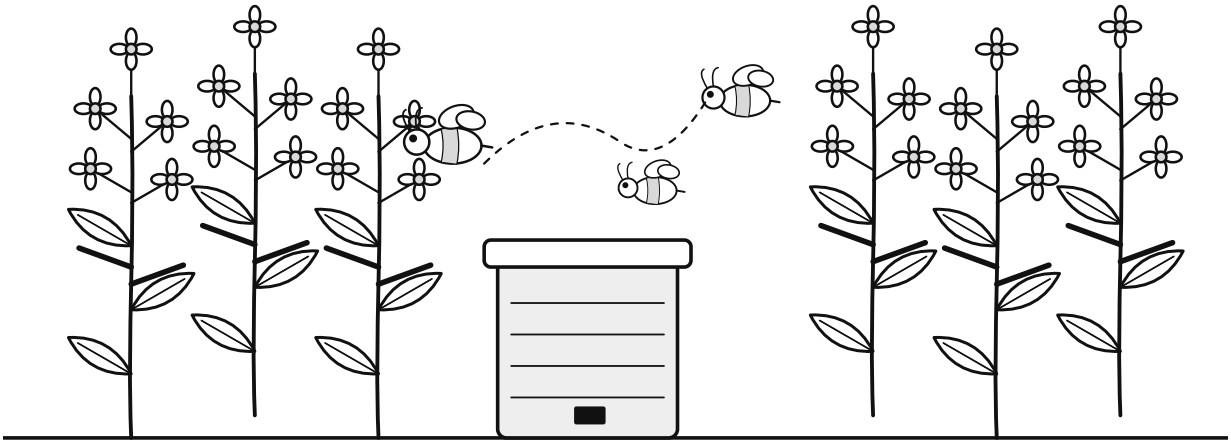
★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Explain how plants and animals depend on each other.
- Use data to compare results.
- Consider different points of view and suggest action.

Science | Maths | Geography | Economics | Ethics



bee box (hive)

From November to January, much of northern and central Bangladesh turns bright yellow with mustard flowers. Farmers grow mustard for its seeds, which are pressed to make **mustard oil** for cooking.

Ayesha is a beekeeper from Sirajganj. Every winter she moves 40 wooden bee boxes to the edge of the mustard fields. "My bees collect nectar from the flowers and turn it into honey," she says. "The farmers like it too, because their mustard plants make more seeds when bees visit."

But there is a problem. Some farmers spray **pesticides** on their crops to kill pests. If the spray is used while the flowers are open and bees are flying, it can kill the bees too. Ayesha now talks to farmers so they spray in the late evening, when bees are back in their hives, or use other ways to control pests.

This story is based on real beekeeping in Bangladesh; the names and numbers are examples for learning.

1 **Explain how the bees and the mustard plants depend on each other.** [2]

2 **Farmer Karim grew mustard in two fields of the same size, with the same seeds, soil and water. Field A was next to Ayesha's bee boxes. Field B was far away from any bees.** [4]

	Field A (near bees)	Field B (no bees nearby)
Mustard seed harvested (kg)	180	150

a) How many more kilograms of seed did Field A produce?

.....

b) Suggest a scientific reason for the difference.

.....

.....

c) Why was it important that the fields were the same size, with the same seeds, soil and water?

.....

.....

3 **Different people see the problem in different ways. What might each person say about spraying pesticides?** [3]

Person	What they might think
A farmer with pests on his crop	
Ayesha, the beekeeper	
A family buying mustard oil and honey	

4 **Suggest two actions that could help both farmers and bees.** [2]

.....

.....

.....

Take it further

Maths: if Field A grew 180 kg and oil is about one third of the seed's mass, about how much oil could be pressed? (180 ÷ 3 = 60 kg). **Geography:** find Sirajganj on a map of Bangladesh.

Show what you know

★★★ Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Design a model that copies how an animal disperses seeds.
- Answer test-style questions.

1 **Find these ten words in the grid.**

Y	S	H	N	I	T	U	H	D	K	F
J	E	A	E	P	O	L	L	E	N	O
A	E	N	C	B	L	Y	S	C	J	W
S	D	T	T	K	A	K	E	J	K	E
T	V	H	A	O	N	U	P	W	F	C
I	J	E	R	U	V	G	A	O	R	O
G	J	R	P	E	I	A	L	M	U	V
M	U	F	K	E	T	A	R	L	I	U
A	B	O	F	L	T	L	J	Y	T	L
T	D	O	G	N	G	A	D	B	B	E
B	F	U	W	E	U	B	L	S	P	T

anther fruit nectar ovary ovule petal pollen seed sepal stigma

2 **Design challenge. Burrs stick to animal fur so their seeds travel. Design a model seed that will stick to a woollen sock and stay on while you walk 10 steps.** [4]

Draw and label your design

a) What materials will you use?

.....

.....

b) How will you test if it works?

.....

.....

.....

3 **Which part of a flower becomes the fruit?** [1]

- ◯ petal
- ◯ ovary
- ◯ stigma
- ◯ anther

4 **A coconut can float across the sea. Which way is it dispersed?** [1]

- ◯ wind
- ◯ water
- ◯ animals
- ◯ explosion

5 **Rashid says, "Seeds need light to make a flower." Explain why he is wrong and what flowers are really for.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name the male and female parts of a flower and say what they do	◯	◯	◯
explain how insects and the wind pollinate flowers	◯	◯	◯
describe what happens at fertilisation	◯	◯	◯
explain why seeds are dispersed and give five ways it happens	◯	◯	◯
put the stages of a flowering plant's life cycle in order	◯	◯	◯
plan a fair test and spot a result that does not fit	◯	◯	◯
explain how plants and animals depend on each other	◯	◯	◯